

Gannaju Sampath Chary

<http://localhost:3000> | RAISE Story |
RAISE App | Projects UI

Python AI Engineer | Data Scientist | Computer Vision | FastAPI |
TensorFlow | PostgreSQL

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PROFESSIONAL SUMMARY

Python AI Engineer with 1.10+ years of hands-on experience building machine learning, computer vision, mobile, web, and backend systems. Experienced across Python, TypeScript, SQL, TensorFlow, PyTorch, OpenCV, FastAPI, Spring Boot, PostgreSQL, TensorFlow Lite, and ONNX Runtime, with practical ownership across data preparation, exploratory analysis, feature engineering, model training, evaluation, deployment, and production support.

CORE COMPETENCIES

AI / Machine Learning	Backend & APIs
Computer Vision, model training, deployment, optimization	Spring Boot, FastAPI, REST APIs, microservices
Full Stack	Data & Delivery
Flutter, React.js, Next.js, UI implementation	PostgreSQL, dataset prep, release support

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Java, Dart
AI / ML	TensorFlow, PyTorch, Scikit-learn, Keras, ONNX Runtime, TensorFlow Lite, OpenCV
Data Science	Pandas, NumPy, Exploratory Data Analysis, Feature Engineering, Statistics, Model Evaluation
Backend & APIs	FastAPI, Spring Boot, REST APIs, Microservices
Frontend & Apps	Flutter, React.js, Next.js, HTML5, CSS3
Database & Delivery	PostgreSQL, MySQL, Firebase, Docker, Google Colab, Google Play Console
Tools	Git, GitHub, Postman, VS Code, Linux

PROFESSIONAL EXPERIENCE

Young Professional - I (AI & Full Stack Developer) ICAR - Indian Institute of Rice Research (IIRR), Hyderabad December 2024 - Present	AI-powered mobile, web, and backend applications for agricultural research.
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- Develop and maintain Python-based AI, web, and backend applications for agricultural research workflows.
- Build production mobile experiences with Flutter and web experiences with React.js and Next.js.
- Design FastAPI and Spring Boot services for authentication, predictions, feedback, and workflow orchestration.

- Collect, clean, preprocess, annotate, and augment domain datasets for model development and validation.
- Perform exploratory data analysis, feature engineering, and dataset quality checks before training.
- Train, validate, and optimize deep learning models for image classification and computer vision use cases.
- Integrate AI models using TensorFlow, PyTorch, TFLite, and ONNX Runtime for mobile and server inference.
- Design PostgreSQL data models and role-based workflows for scientists, admins, and end users.
- Collaborate with subject-matter experts for ground-truth validation, evaluation, and iteration.
- Deploy applications and backend services and support production releases and debugging.

FLAGSHIP PROJECT - RAISE	
RAISE Rice AI Stress Evaluator Project Story Open App	Flutter TensorFlow Spring Boot Python PostgreSQL TFLite ONNX Runtime AI-powered image-based platform for identifying biotic and abiotic rice crop stresses, combining mobile inference, server-side AI services, scientist validation, and structured agricultural data management.

- Developed the Flutter Android application with image capture, crop-stress scanning, prediction history, offline inference, multilingual support, and farmer-focused workflows.
- Built the React.js scientist/admin portal and Spring Boot backend with REST APIs for authentication, predictions, expert review, data management, and administration.
- Integrated FastAPI-based Python AI services with TensorFlow/PyTorch models for server-side image inference.
- Trained and evaluated image-classification models including EfficientNetV2, EfficientNetV1, ResNet, MobileNet, and InceptionNet; explored YOLOv5 and YOLOv8 for computer-vision workflows.
- Converted and optimized models for lightweight inference using TensorFlow Lite and ONNX Runtime.
- Implemented Farmer, Scientist, and Admin roles, including expert review for visually confusing cases and feedback-driven model improvement.
- Designed PostgreSQL data models for users, predictions, review workflows, and application history.

PERSONAL PROJECTS

RAISE

Rice AI Stress Evaluator
Project Story

Flutter | TensorFlow | Spring Boot | Python | PostgreSQL | TFLite | ONNX
Runtime
Open live experience
No backend link

Production AI platform for identifying biotic and abiotic rice crop stresses using image-based diagnosis. Includes mobile AI inference, scientist/admin portal, expert validation, role-based access, prediction history, multilingual UI, and offline inference.

VistaraAI

AI-powered PVC interior design assistant
Project Story

Next.js | TypeScript | React | FastAPI | Python | LangChain | LangGraph | PostgreSQL
Open live experience
Backend API

Production-ready full-stack AI platform for PVC kitchen cupboards, wardrobes, TV units, and custom interiors. Combines grounded knowledge retrieval, assistant workflows, computer vision insights, geo-targeted SEO, and lead capture support.

RiceGPT AI

Agricultural AI assistant concept
Project Story

Next.js | TypeScript | React
Open live experience
Backend API

Domain AI assistant concept for agricultural and rice-related queries, pairing a structured knowledge base with a public web deployment.

ResumeAI Pro

AI resume optimization concept
Project Story

Next.js | TypeScript | React | Tailwind CSS | FastAPI
Open live experience
Backend API

AI-assisted resume analysis and refinement concept focused on recruiter-friendly guidance and practical deployment.

Interview Copilot AI

Interview preparation companion
Project Story

React | TypeScript | Prompt Engineering
Open live experience
Backend API

Interview preparation companion for technical question practice, answer structuring, and confidence building.

DataInsight AI

AI data analysis concept
Project Story

Python | Next.js | TypeScript | Data Visualization
Open live experience
Backend API

AI-assisted exploratory data analysis concept for pattern discovery, summary reporting, and decision support.

AutoML Studio

Automated ML workflow concept
Project Story

Python | FastAPI | Scikit-learn | Next.js | Gemini
Open live experience
Backend API

End-to-end automated machine learning concept for dataset upload, preprocessing, training, comparison, and evaluation.

ADDITIONAL PROJECT EXPERIENCE**Rice Clinic**

Flutter | Localization | Image Matching | Google Apps
Script | Google Sheets

Multilingual rice advisory application for diseases, pests, weeds, nutrient deficiencies, cultivation practices, and fertilizer/spray recommendations.

KEY ACHIEVEMENTS

- Delivered an end-to-end AI application spanning dataset preparation, model training, mobile/web development, backend APIs, database design, deployment, and maintenance.
- Published the RAISE IIRR Android application on the Google Play Store.
- Built a multi-role AI platform connecting farmers, scientists, administrators, and expert reviewers through a single application ecosystem.
- Developed model-deployment workflows supporting both offline mobile inference and online server inference.
- Built reusable backend and AI-service architecture using Spring Boot, FastAPI, PostgreSQL, and Python.

EDUCATION**B.Tech, Computer Science**

2019 - 2023

Gurunanak Institutions Technical Campus, Hyderabad,
Telangana

University not explicitly specified in the current resume source.

Portfolio: <http://localhost:3000> | Projects: </projects> | Experience: </experience> | Resume: </resume/resume.pdf>